PEER REVIEW HISTORY

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ARTICLE DETAILS

TITLE (PROVISIONAL)	An updated systematic review and meta-analysis of studies examining the relationship between reported racism and health and wellbeing for children and youth: A protocol
AUTHORS	Priest, Naomi; Doery, Kate; Truong, Mandy; Guo, Shuaijun; Perry, Ryan; Trenerry, Brigid; Karlsen, Saffron; Kelly, Yvonne; Paradies, Yin

VERSION 1 – REVIEW

REVIEWER	Leah Cave Telethon Kids Institute
REVIEW RETURNED	07-Oct-2020

GENERAL COMMENTS	Thank you for the opportunity to review this review protocol. The protocol presents a comprehensive update of an earlier systematic review of studies investigating associations between racial discrimination and health outcomes in children and adolescents 0-24 years - expanding on this previous review with meta-analysis, a wider age range for participants, and broader health outcome criteria. This review will provide a powerful contribution to the review literature and further our understanding of the health effects of racial discrimination early in the life course. Some very minor typos were present which have been highlighted in the attached pdf proof (pages 2, 3 & 7). The reviewer provided a marked copy with additional comments. Please contact the publisher for full details.

REVIEWER	Uday N Yadav UNSW, Sydney, Australia
REVIEW RETURNED	01-Nov-2020

GENERAL COMMENTS	This is really a good piece of work. However, I do have some queries and suggestions. I would like to know more about exposure measures and readers would like to know if you are including both interpersonal and systemic racism. -I would like to know in which primary outcomes you are going to include Violence and bullying parameters. -Exclusion criteria requires some more info like you are not including editorials, perspectives, RCT, CCT etc. - What about COVID-19 racism and children health outcomes? I could not find the search terms in this line. As authors have continuously mentioned that racism is increased in COVID era in their paper, I would suggest the research team to look at this issue.

	Just for your read: https://pediatrics.aappublications.org/content/early/2020/10/28/peds.2020-021816 - I would suggest adding key terms for adolescents from indigenous/marginalised communities as well because they are the one who experiences racism more than anyone. It will definitely increase the value of this work. You shall have a look at our work: Kairuz, C.A., Casanelia, L.M., Bennett-Brook, K. et al. Impact of racism and discrimination on the physical and mental health among Aboriginal and Torres Strait Islander peoples living in Australia: a protocol for a scoping review. Syst Rev 9, 223 (2020). https://doi.org/10.1186/s13643-020-01480-w In the discussion section, the authors need to provide how the findings will be useful for policy and future interventions for the parameters that authors are measuring in this review.

REVIEWER	Emily Tweed University of Glasgow, UK
REVIEW RETURNED	05-Jan-2021

GENERAL COMMENTS	Thank you for the opportunity to review this interesting submission. This protocol describes a planned update to a previous review carried out by Priest and colleagues (2013), which found an association between reported racial discrimination and poor health among children and adolescents. This review aims to update and extend the previous review, notably including a wider range of ages (up to 24 years rather than 18 years as previously) and undertaking meta-analysis. It is timely given the increased attention to structural racism in recent years in wider society and in particular in relation to health. The authors provide appropriate justification for the decision to update the previous review based on the growing attention to racism as a social determinant of health including new pathophysiological evidence and a focus on vicarious racism. The review is very broad in scope in that it will potentially retrieve a substantial body of papers to be reviewed (based on the previous review and publication trends since) and the authors plan to extract a substantial volume of data from each one. This is an important topic and the review is to be welcomed. There are several ways in which I feel the protocol needs to be strengthened prior to publication, as described below: these are primarily requests for more clarity or detail on a number of important topics, rather than suggestions of how the approach should be changed. The main methodological issue which I feel does need to be reviewed is the approach to assessment of risk of bias, as detailed below. Comments are roughly grouped below according to the review checklist. Is the research question or study objective clearly defined? The title of the paper as submitted does not identify it as a protocol for a systematic review or an update of a previous review which requires corrected (as per PRISMA-P checklist). Another administrative point as per PRISMA-P – the guarantor of the review is not identified. The introduction describes two “broad aims” for the review but does not specify a clearly defined research question – this is an important omission which I feel should be addressed. As per PRISMA-P the question should be provided and include reference

	to participants, interventions (or exposures, in this case), comparators, and outcomes Given that the age range now extends up to 24 years, would it be more appropriate to refer in the title and text to “children and young people” rather than “children and adolescents”? There is a subsequent justification for an extended definition of this term with reference to a ‘Viewpoint’ article: this isn’t my specialist area so I wasn’t sure whether this single article was sufficient basis for this choice. With regard to the description of the context in the introduction, there is a reference to the simultaneous pandemics of Covid-19 and racism – I think this would benefit from being explained and justified more clearly, as it is not clear from what is written at present whether the implied claim of a rapid recent upward trend in racism (as opposed to, for instance, long-standing endemic racism receiving greater attention through e.g. BLM movements) is justified. 2. Is the abstract accurate, balanced and complete? The abstract describes the context of increasing research in this area and describes the methods in terms of scope of population, exposure, outcome and in terms of how studies will be screened and extracted. Given the focus on meta-analysis it may be worth briefly alluding to this in the methods. I also felt that the dissemination section was rather vague and would benefit from some more concrete material. The abstract refers to “racial dissemination” in line 44, which I presume is a typo and instead should read racial discrimination? 3. Is the study design appropriate to answer the research question? The inclusion criteria refer to the inclusion of grey literature but this seems to be missing from the search strategy, other than the use of ProQuest for dissertation/theses and a vague reference to searching “key websites” and contacting experts. I would welcome more information on how the grey literature will be searched e.g. which websites will be included. The authors refer to hand-searching reference lists – it wasn’t obvious to me (though may be to others!) whether this was only for the reference lists of included studies; I presume so given that even this will be a substantial undertaking, but it would be good to spell this out. The authors appropriately acknowledge that restricting their search to English is an important limitation. The time period is a strength given the authors’ acknowledgement that studies are frequently back-indexed and may have been missed by the previous review. The section on proof page 9/27 from lines 27-45 is somewhat unclear in purpose – I initially thought it was describing what data was to be extracted but this is specified in the following paragraphs, so instead it reads more like a description of the research questions (“this review will examine”). I felt it would be good to clarify what is being described here – perhaps by integrating this with the description of what data will be extracted and how it will be synthesised. I was not convinced about the approach to choosing critical appraisal tools as the description provided seemed somewhat ad-hoc and potentially lacking in replicability/transparency, but this is not my area of expertise. From my inexpert perspective, I wonder whether it would not be preferable to use the Systematic Review Toolbox mentioned to identify the most appropriate tools for the

	limited set of study types included in advance, so they can be specified in the protocol? Moreover, I was somewhat confused by the reference to ROBINS-E in the context of publication bias. ROBINS-E is a tool for assessing the risk of bias based on the way the study was carried out, as a means of understanding validity of results, NOT a tool for assessing the risk of publication bias (i.e. differential publication of studies based on their results). ROBINS-E is more appropriate for the critical appraisal process described in the preceding paragraph. The authors also include in the appendix a preliminary tool used in the development of ROBINS-E rather than ROBINS-E tool itself (still under development), without acknowledging this preliminary status, and cite in support of their use of this tool a paper which makes a number of criticisms of its design and use. I would therefore suggest that the section on quality assessment and publication bias needs substantial revision and clarification. Again, the description of the methods in the body of the text would benefit from stating explicitly whether assessment of study quality will be undertaken independently by two separate members of the team (as per the abstract). There is no information provided on how the assessment of risk of bias will be used in data synthesis, other than a vague reference as “used as a guide for data analysis and meta-analysis”. I would welcome more information on this e.g. will assessment of RoB only be used as part of the narrative description of studies or will it be used to e.g. stratify meta-analysis. Specifying how this information will be used in synthesis is a requirement of the PRISMA-P checklist: at present I don’t feel there is enough information to meet this requirement. On proof page 9/27 line 8 there is a reference to studies not suitable for meta-analysis being documented but excluded from the quantitative analysis, which – after reading the subsequent section on analysis – seems potentially misleading to the reader given the plans for substantial narrative analysis of studies prior to meta-analysis (that is, it underplays the role these studies will play in the final review results). I would suggest this sentence could be dropped or amended to reflect this. In general I would welcome more details on the narrative synthesis, beyond just a reference to “a descriptive summary” - the SWIM (synthesis without meta-analysis) reporting guideline may be useful here. I feel this is important given that the scope and diversity of potential data points included means that meta-analysis may only be feasible for a proportion – and given the fact that contextual factors are likely to be extremely important to the relationship under study. A clearer description of plans for narrative synthesis would ensure the authors are able to utilise to best effect the large quantities of data they plan to extract from each study. There is also no details on the final PRISMA-P requirement, how the strength of the body of evidence will be assessed overall. 13. Is the supplementary reporting complete (e.g. trial registration; funding details; CONSORT, STROBE or PRISMA checklist)? The PRISMA-P checklist is included but a number of items are missing from the report as described above. 4. Are the methods described sufficiently to allow the study to be repeated? I would welcome more information on how the authors conceptualise racism within the scope of the review in order to

	implement the inclusion criteria relating to exposure. This is especially important given the exclusion criteria referring to "studies where the specific effects of racism cannot be isolated from broader measures of discrimination" – this seems to me to require a working definition of racism in order to implement this criterion, so it would be good to have this spelled out in the protocol. The description of data extraction in the body of the text is slightly unclear about whether data extraction will be undertaken independently by two separate members of the team – this is stated in the abstract but is not obvious from the text on proof page 9/27, lines 14-19, so it would be gaood just to tighten up the wording of this section to ensure this is clearly stated. As a minor point, the PROSPERO ID is mentioned at the end of the protocol under ethics and dissemination but it would also be good to include this at proof page 6/27 line 35 when PROSPERO is first mentioned. The section on dissemination would benefit from being more detailed and concrete, as at the present it is rather vague and generic.

VERSION 1 – AUTHOR RESPONSE

Reviewers' Comments	**Authors' Response**	**Reference page**
Reviewer 1- Dr. Leah Cave, Telethon Kids Institute		
Thank you for the opportunity to review this review protocol. The protocol presents a comprehensive update of an earlier systematic review of studies investigating associations between racial discrimination and health outcomes in children and adolescents 0-24 years - expanding on this previous review with meta-analysis, a wider age range for participants, and broader health outcome criteria. This review will provide a powerful contribution to the review literature and further our understanding of the health effects of racial discrimination early in the life course.	We thank the reviewer for their comments.	
Some very minor typos were present which have been highlighted in the attached pdf proof (pages 2, 3 & 7).	The typos have been updated.	-
Reviewer 2- Dr. Uday Yadav, University of New South Wales		
I would like to know more about exposure measures and readers would like to know if you are including both interpersonal and systemic racism.	Studies examining interpersonal and systemic racism exposure measures will be included if all other study inclusion criteria are also met. In order to clarify the exposure measures of interest in the review, we have added a	Page 2, 4

Reviewers' Comments	Authors' Response	Reference page
	definition of each of these (systemic, interpersonal and internalised racism) in the introduction. We have now also expanded the methods to clarify the exposure types in line with this definition. "This review will extract childhood experiences of racism and racial discrimination exposures, synthesised across three main categories in the meta-analysis: systemic racism; interpersonal experiences including direct and vicarious or second hand experiences (for example, witnessing or hearing about racism experienced by family, friends or a group an individual belongs to); and internalised racism (for example, adapting racist attitudes and/or beliefs within their world views). Both personal and proxy reports (by parents or caregivers) of experiences of racism will be recorded, as will the source of racism (categorised as peers, teachers, and community). Terms used to determine the exposure include but are not limited to racism, discrimination, prejudice, harassment, bullying, stereotypes and unfair treatment where reason includes the victims' race/ethnicity/cultural background or proxy indicators such as migrant background, skin colour, language or accent. Although we recognise that religion is often highly racialised, studies of religious discrimination will not be included"	
I would like to know in which primary outcomes you are going to include Violence and bullying parameters.	Thank you for raising this point for clarification. Studies assessing violence as an outcome is associated with experiences of racism will be categorised as "Negative mental health" as this outcome category includes in its definition, "antisocial behaviours including aggression and violence." O'Connor M, Sanson A, Hawkins MT, et al. Differentiating three conceptualisations of	Page 5

Reviewers' Comments	Authors' Response	Reference page
	the relationship between positive development and psychopathology during the transition to adulthood. Journal of adolescence 2011;34(3):475-84. Bullying is not seen as a mental health impact and is seen more as a risk factor and therefore will not be included as a health outcome for this review as per the following citations. Scott JG, Moore SE, Sly PD, Norman RE. Bullying in children and adolescents: a modifiable risk factor for mental illness. Australian & New Zealand Journal of Psychiatry. 2014 Mar;48(3):209-12. Fisher HL, Moffitt TE, Houts RM, Belsky DW, Arseneault L, Caspi A. Bullying victimisation and risk of self harm in early adolescence: longitudinal cohort study. Bmj. 2012 Apr 26;344:e2683.	
Exclusion criteria requires some more info like you are not including editorials, perspectives, RCT, CCT etc.	Our inclusion criteria states that only primary empirical studies will be included. We have reiterated this to state that editorials and commentaries will not be included unless they report primary empirical data. As noted in the methods this study will include primary empirical studies that use quantitative designs. We now explicitly note that RCTs and CCTs will be included. Based on our knowledge of the field we expect included studies to be largely observational cross-sectional and cohort studies. "This report will include primary empirical studies that use quantitative methods including but not limited to cross-sectional; prospective and retrospective cohort; case–control designs; quasi-experimental studies and randomised control trials. Peer-reviewed journal articles (published or	Page 4

Reviewers' Comments	Authors' Response	Reference page
	available as pre-prints), and dissertations/theses will be included. We will also include grey literature such as published reports. Studies that do not report primary empirical associations between racism and child and youth health will not be included. Editorials and commentaries will not be included unless they report primary empirical data. According to the PICO (Population, Intervention, Comparison and Outcomes) tool outlined in the Cochrane Handbook,[20] we defined our population, intervention (exposure of interest) and specific outcome measures. "	
What about COVID-19 racism and children health outcomes? I could not find the search terms in this line. As authors have continuously mentioned that racism is increased in COVID era in their paper, I would suggest the research team to look at this issue. Just for your read: https://pediatrics.aappublications.org/content/early/2020/10/28/peds.2020-021816	Thank you for this comment. We agree COVID-19 has heightened racism experiences for many children and youth. However, this is not a primary focus of the review, that is, the extent to which COVID-19 influences the prevalence of racism for children or the extent to which the pandemic moderates the health effects or racism, while important questions are not the focus of this review. References to COVID have been removed from the paper. The search strategy was developed in consultation with expert medical librarians and we are confident that studies on COVID-19 racism and child health will be identified by the current search strategy which is very broad. It is not necessary to include COVID-19 as a specific search term as any study that includes racism and children and young people will already be captured by the present search strategy and denoting specific contexts such as COVID-19 is not required and would only narrow the search.	-
I would suggest adding key terms for adolescents from indigenous/marginalised communities as well because they are the one who experiences racism more than	Thank you for this suggestion. Again, as with the response above regarding COVID, we are confident that our search strategy will capture all relevant studies focussing	-

Reviewers' Comments	Authors' Response	Reference page
anyone. It will definitely increase the value of this work. You shall have a look at our work: Kairuz, C.A., Casanelia, L.M., Bennett-Brook, K. et al. Impact of racism and discrimination on the physical and mental health among Aboriginal and Torres Strait Islander peoples living in Australia: a protocol for a scoping review. Syst Rev 9, 223 (2020). https://doi.org/10.1186/s13643-020-01480-w	on indigenous and marginalised communities. Adding additional search terms related to Indigenous or marginalised communities will only narrow rather than widen the search strategy. As such it is unnecessary.	
In the discussion section, the authors need to provide how the findings will be useful for policy and future interventions for the parameters that authors are measuring in this review.	We agree that policy and intervention implications of this review are important. The original systematic review has already had considerable policy and practice impact. As such we have added to the discussion "Review findings will provide essential information for future research, policy priorities and inform the development effective evidenced based targets for interventions to ameliorate the harmful impacts of racism on child and adolescent health."	Page 9
Reviewer 3- Dr. Emily Tweed, University of Glasgow		
Thank you for the opportunity to review this interesting submission. This protocol describes a planned update to a previous review carried out by Priest and colleagues (2013), which found an association between reported racial discrimination and poor health among children and adolescents. This review aims to update and extend the previous review, notably including a wider range of ages (up to 24 years rather than 18 years as previously) and undertaking meta-analysis. It is timely given the increased attention to structural racism in recent years in wider society and in particular in relation to health. The authors provide appropriate justification for the decision to update the previous review based on the growing attention to racism as a social determinant of health including new pathophysiological evidence and a focus on vicarious racism. The review is very broad in scope in that it will potentially retrieve a substantial body of papers to be reviewed (based on the previous review and publication trends since) and the authors plan to extract a substantial volume of data from each one. This is an important topic and the review is to be welcomed. There are several ways in which I feel the protocol needs to be	We thank the reviewer for their comments. We have addressed their comments as detailed below.	-

Reviewers' Comments	Authors' Response	Reference page
strengthened prior to publication, as described below: these are primarily requests for more clarity or detail on a number of important topics, rather than suggestions of how the approach should be changed. The main methodological issue which I feel does need to be reviewed is the approach to assessment of risk of bias, as detailed below. Comments are roughly grouped below according to the review checklist		
1. Title, guarantor, research question and study population		
The title of the paper as submitted does not identify it as a protocol for a systematic review or an update of a previous review which requires corrected (as per PRISMA-P checklist). Another administrative point as per PRISMA-P – the guarantor of the review is not identified.	The title has now been updated to state this is an updated protocol. **"An updated systematic review and meta-analysis of studies examining the relationship between reported racism and health and wellbeing for children and youth: a protocol."** The first author as the guarantor of the review is now noted in the author contributions. "NP conceptualised the review, contributed to all aspects of the protocol and the guarantor for the review"	Title page Author contributions
The introduction describes two "broad aims" for the review but does not specify a clearly defined research question – this is an important omission which I feel should be addressed. As per PRISMA-P the question should be provided and include reference to participants, interventions (or exposures, in this case), comparators, and outcomes	We have now framed the broad aims of the review as a question to be consistent with PRISMA-P: "To what extent are experiences of racism associated with health and wellbeing outcomes among children and adolescents compared to those who experience no or less racism?" This question reference participants (children and adolescents), exposure (racism), and outcomes (health and wellbeing).	Page 4
Given that the age range now extends up to 24 years, would it be more appropriate to refer in the title and text to "children and young people" rather than "children and adolescents"? There is a subsequent	This has now been updated and youth is now used throughout. An additional reference has been added to highlighted that the ages 10-24 are used by key	Page 4

Reviewers' Comments	Authors' Response	Reference page
justification for an extended definition of this term with reference to a 'Viewpoint' article: this isn't my specialist area so I wasn't sure whether this single article was sufficient basis for this choice.	organisations such as the Lancet Commission on adolescent health. "The age range of participants has been updated since the previous review (which included participants up to 18 years) to include children and youth as per the United Nations definition.[21-23] This broader age range will allow for consideration of the health impacts of racism into late adolescence which is now considered by the Lancet Commission on Adolescent Health as extending to 24 years.[23] Sawyer SM, Afifi RA, Bearinger LH, et al. Adolescence: A foundation for future health. *The Lancet* 2012;379(9826):1630-40. doi: 10.1016/S0140-6736(12)60072-5 United Nations. Youth: United Nations; 2020 [Available from: https://www.un.org/en/sections/issues-depth/youth-0/index.html2020. Patton GC, Sawyer SM, Santelli JS, et al. Our future: a Lancet commission on adolescent health and wellbeing. *The Lancet* 2016;387(10036):2423-78.	
With regard to the description of the context in the introduction, there is a reference to the simultaneous pandemics of Covid-19 and racism – I think this would benefit from being explained and justified more clearly, as it is not clear from what is written at present whether the implied claim of a rapid recent upward trend in racism (as opposed to, for instance, long-standing endemic racism receiving greater attention through e.g. BLM movements) is justified.	We have removed reference to COVID-19 as this is not a primary aim of the review and was distracting.	-
2. Is the abstract accurate, balanced and complete?		

Reviewers' Comments	Authors' Response	Reference page
The abstract describes the context of increasing research in this area and describes the methods in terms of scope of population, exposure, outcome and in terms of how studies will be screened and extracted. Given the focus on meta-analysis it may be worth briefly alluding to this in the methods. I also felt that the dissemination section was rather vague and would benefit from some more concrete material.	We have now included two sentences in this section of the abstract specifically identifying the meta-analysis approach. The dissemination section has also been strengthened. "A random-effects meta-analysis will examine the average effect of racism on a range of health outcomes. Study-level moderation will test the difference in effect sizes with regard to various sample and exposure characteristics." "Specific dissemination strategies will be co-designed with policy, practice and community stakeholders towards completion of the review."	Abstract; Page 9
The abstract refers to "racial dissemination" in line 44, which I presume is a typo and instead should read racial discrimination?	Updated	Page 1
3. Is the study design appropriate to answer the research question?		
The inclusion criteria refer to the inclusion of grey literature but this seems to be missing from the search strategy, other than the use of ProQuest for dissertation/theses and a vague reference to searching "key websites" and contacting experts. I would welcome more information on how the grey literature will be searched e.g. which websites will be included.	To clarify the intended approach for searching grey literature, we have updated the search strategy to state "The search will be conducted in the Ovid Medline, Ovid PsycInfo, PubMed, ERIC and ProQuest (for dissertation/theses) databases. Reference lists of included studies will also be hand-searched for additional relevant studies. The authors will also search google scholar to identify papers and reports citing the previous review and will search grey literature databases including Open Grey, OpenDOAR and New York Academy of Medicine using keywords from the search strategy."	Page 6
The authors refer to hand-searching reference lists – it wasn't obvious to me (though may be to others!) whether this was only for the reference lists of included studies; I presume so given that even this will be a substantial undertaking, but it would be good to spell this out.	Yes- Reference list searches will be conducted of included studies. This has been clarified in text.	Page 6

Reviewers' Comments	Authors' Response	Reference page
	"Reference lists of included studies will also be hand-searched for additional relevant studies."	
The authors appropriately acknowledge that restricting their search to English is an important limitation.	Thank you.	-
The time period is a strength given the authors' acknowledgement that studies are frequently back-indexed and may have been missed by the previous review.	Thank you.	-
The section on proof page 9/27 from lines 27-45 is somewhat unclear in purpose – I initially thought it was describing what data was to be extracted but this is specified in the following paragraphs, so instead it reads more like a description of the research questions ("this review will examine"). I felt it would be good to clarify what is being described here – perhaps by integrating this with the description of what data will be extracted and how it will be synthesised.	These dot points have been removed and integrated into a now expanded description of the exposure data that will be extracted and synthesised in the review and meta-analysis.	-
I was not convinced about the approach to choosing critical appraisal tools as the description provided seemed somewhat ad-hoc and potentially lacking in replicability/transparency, but this is not my area of expertise. From my inexpert perspective, I wonder whether it would not be preferable to use the Systematic Review Toolbox mentioned to identify the most appropriate tools for the limited set of study types included in advance, so they can be specified in the protocol? Moreover, I was somewhat confused by the reference to ROBINS-E in the context of publication bias. ROBINS-E is a tool for assessing the risk of bias based on the way the study was carried out, as a means of understanding validity of results, NOT a tool for assessing the risk of publication bias (i.e. differential publication of studies based on their results). ROBINS-E is more appropriate for the critical appraisal process described in the preceding paragraph. The authors also include in the appendix a preliminary tool used in the development of ROBINS-E rather than ROBINS-E tool itself (still under development), without acknowledging this preliminary status, and cite in support of their use of this tool a paper which makes a number of criticisms of its design and use. I would therefore suggest that the section on quality assessment and publication bias needs substantial revision and clarification.	We have revised the first paragraph in the section on assessment of study quality to clearly articulate our process for selecting and interpreting an appropriate quality assessment tool. Consistent with the reviewer's suggestion here, we used the Systematic Review Toolbox to select the Newcastle-Ottawa Scale as our quality appraisal tool. We no longer reference any other quality appraisal option in the protocol. We have also included a new paragraph in the section briefly detailing the publication bias tests (i.e., Egger and trim-and-fill) that we will use. We no longer refer to ROBINS-E in the protocol. "Studies included in the review will be critically appraised to determine the validity of the study's findings from the known literature and to provide readers with the ability to make an informed decision on the quality of these findings. Two members of the review team will independently rate studies selected for the meta-analysis	Page 7

Reviewers' Comments	Authors' Response	Reference page
	using the Newcastle-Ottawa Scale, a widely used tool for evaluating the quality of non-randomised studies on a range of criteria.[31] The Newcastle-Ottawa Scale was determined to be the most appropriate instrument for quality assessment via a consideration of tools available in the Systematic Review Toolbox.[32] Any discrepancies between quality ratings will be resolved through discussion with the lead author. The quality ratings for each study will be provided in a table in the supplementary material and studies with scores deemed less than satisfactory will be discussed as part of the narrative analysis. We will not exclude any articles from analysis due to low quality scores, but sensitivity analysis will be performed to determine any effect of including studies that score less than "good" according to Agency for Health Research and Quality (AHRQ) standards.[33] Evidence of publication bias and small-study effects will be assessed using two methods (available in the Stata meta suite).[34] First, we will examine contour-enhanced funnel plots for asymmetry using the Egger regression test.[35] Second, we will use trim-and-fill analysis to estimate effects sizes that are adjusted for publication bias.[36]"	
Again, the description of the methods in the body of the text would benefit from stating explicitly whether assessment of study quality will be undertaken independently by two separate members of the team (as per the abstract).	This has been updated to state that two reviewers will complete data extraction and will resolve any discrepancies by discussion with the lead author. "Two reviewers will independently extract data from each study, with inconsistencies and discrepancies resolved through discussion."	Page 6
There is no information provided on how the assessment of risk of bias will be used in data synthesis, other than a vague reference as "used as a guide for data	As described above we have substantially revised the section of the protocol titled "Assessment of study quality and bias" to	Page 7

Reviewers' Comments	Authors' Response	Reference page
analysis and meta-analysis". I would welcome more information on this e.g. will assessment of RoB only be used as part of the narrative description of studies or will it be used to e.g. stratify meta-analysis. Specifying how this information will be used in synthesis is a requirement of the PRISMA-P checklist: at present I don't feel there is enough information to meet this requirement.	clearly describe our intended processes for assessing risk of bias. As noted in this section, we will assess risk of bias by scoring studies on quality according to the Newcastle-Ottawa Scale. We also note here that studies deemed to be at higher risk of bias (i.e., with low score on the Newcastle-Ottawa) will be detailed as part of the narrative analysis and sensitivity analyses will be performed to determine whether inclusion of these at-risk studies has any effect on the meta-analysis results.	
On proof page 9/27 line 8 there is a reference to studies not suitable for meta-analysis being documented but excluded from the quantitative analysis, which – after reading the subsequent section on analysis – seems potentially misleading to the reader given the plans for substantial narrative analysis of studies prior to meta-analysis (that is, it underplays the role these studies will play in the final review results). I would suggest this sentence could be dropped or amended to reflect this.	This line has been removed as per the reviewer's suggestion.	-
In general I would welcome more details on the narrative synthesis, beyond just a reference to "a descriptive summary" - the SWIM (synthesis without meta-analysis) reporting guideline may be useful here. I feel this is important given that the scope and diversity of potential data points included means that meta-analysis may only be feasible for a proportion – and given the fact that contextual factors are likely to be extremely important to the relationship under study. A clearer description of plans for narrative synthesis would ensure the authors are able to utilise to best effect the large quantities of data they plan to extract from each study.	This detail has now been expanded to highlight that the narrative synthesis will follow the original review and also draw on the SWIM guidelines (noting these guidelines are for intervention studies and not all criteria apply to observational studies). "Following the format of the original review and drawing on the SWiM[37] reporting items, a narrative synthesis of study characteristics and findings will be conducted. This will provide a description and rationale for the reporting of groups used in the synthesis such as study populations, outcomes, study designs, methods used to assess the certainty of the evidence and limitations of the review. Study characteristics will be presented in	Page 8

Reviewers' Comments	Authors' Response	Reference page
	summary tables across key variables (including study design, year, setting, country; population characteristics; exposure measures including number of items and whether validated; definition of racism used. If meta-analysis is not possible the nature of the relationship between exposure and outcome (positive, negative or null) across key study characteristics will be summarised, following the approach in the original review.[1]"	
There is also no details on the final PRISMA-P requirement, how the strength of the body of evidence will be assessed overall.	We will use the Grading of Recommendations, Assessment, Development and Evaluation (GRADE) to summarize the overall evidence according to each outcome. This has now been added to the methods section. "Based on the Grading of Recommendations, Assessment, Development and Evaluation (GRADE) framework,[38] we will rate the quality of the overall evidence across each outcome to conclude."	Page 8
4. Are the methods described sufficiently to allow the study to be repeated?		
I would welcome more information on how the authors conceptualise racism within the scope of the review in order to implement the inclusion criteria relating to exposure. This is especially important given the exclusion criteria referring to "studies where the specific effects of racism cannot be isolated from broader measures of discrimination" – this seems to me to require a working definition of racism in order to implement this criterion, so it would be good to have this spelled out in the protocol.	Thank you. The definition of racism has now been expanded in the introduction and clarity on what will be excluded has been added to the methods. "Racism is expressed across multiple levels, including systemic or structural racism, embedded within society; interpersonal racism between two or more individuals and includes racially motivated assaults or abuse; and internalised racism, whereby people take on negative stereotypes and beliefs about themselves.[6-8] Racism and racial discrimination operates in many forms	Page 2, 4, 5

Reviewers' Comments	Authors' Response	Reference page
	including direct and vicarious racism (second hand racism), whereby individuals experience racism on a secondary level, witnessing or being informed of family, friends and strangers experiencing racism." "Studies that do not attribute discrimination experiences to race or ethnicity but only report generalised discrimination or unfair treatment without attribution will not be included."	
The description of data extraction in the body of the text is slightly unclear about whether data extraction will be undertaken independently by two separate members of the team – this is stated in the abstract but is not obvious from the text on proof page 9/27, lines 14-19, so it would be good just to tighten up the wording of this section to ensure this is clearly stated.	Thank you for this comment. This has now been edited to clarify that two people will do all data extraction.	Page 6
As a minor point, the PROSPERO ID is mentioned at the end of the protocol under ethics and dissemination but it would also be good to include this at proof page 6/27 line 35 when PROSPERO is first mentioned.	This has been included in the Abstract and when PROSPERO was first mentioned.	Abstract; Page 4
The section on dissemination would benefit from being more detailed and concrete, as at the present it is rather vague and generic.	This has now been addressed. "Specific dissemination strategies will be co-designed with policy, practice and community stakeholders towards completion of the review."	Page 9
The PRISMA-P checklist is included but a number of items are missing from the report as described above.	This has now been addressed. See Supplementary File.	Supplementary file
Is the supplementary reporting complete (e.g. trial registration; funding details; CONSORT, STROBE or PRISMA checklist)?	We have added trial registration with PROSPERO, no funding for project, and updated the PRISMA checklist.	Supplementary file

VERSION 2 – REVIEW

REVIEWER	Yadav, Uday University of New South Wales
REVIEW RETURNED	01-Apr-2021

GENERAL COMMENTS	Dear authors, thanks for addressing the remarks and providing me the opportunity to review your piece. Still, I have few suggestions for your team- I would suggest authors to add some sentences highlighting the impact of racism among youths from Indigenous and marginalised communities. Perhaps after line line 52 of page 2. Please have a look and cite it holds significance in this context: https://systematicreviewsjournal.biomedcentral.com/articles/10.1186/s13643-020-01480-w I am wondering if you have any team member of your review team represent truly marginalised/ Indigenous community. I presume your team must be an expert in this area of research but I feel involvement of one or two representative from marginalised/indigenous background would bring great flavour to data interpretation. This is solely based on my learnings.

REVIEWER	Tweed, Emily University of Glasgow, MRC/CSO Social and Public Health Sciences Unit
REVIEW RETURNED	09-Mar-2021

GENERAL COMMENTS	Thank you for your detailed response to the first round of comments; all the areas identified in the first review appear to have been addressed.

VERSION 2 – AUTHOR RESPONSE

I would suggest authors to add some sentences highlighting the impact of racism among youths from Indigenous and marginalised communities. Perhaps after line line 52 of page 2. Please have a look and cite it holds significance in this context:	We have reinforced this point in the final sentence of the first paragraph of the introduction: *Racism profoundly and perniciously impacts indigenous and racialised peoples throughout the world, including children and youth, for whom racism and racial discrimination is a major burden and influence on their health and development throughout life.*[1 2 11 12] We have considered the suggested reference, as we did at the last review, and do not consider it the most appropriate reference to reinforce this point. The suggested citation – which we note is the reviewer's own paper - is a review protocol rather than a completed review or empirical paper – and draws heavily on much of our own theoretical and empirical work. Further, the suggested citation is

	focused on one specific population and context whereas ours is a global review. As such we have cited key global papers in the field to support our point. We query the reviewer's insistence on citing their work in this context.
I am wondering if you have any team member of your review team represent truly marginalised/ Indigenous community. I presume your team must be an expert in this area of research but I feel involvement of one or two representative from marginalised/indigenous background would bring great flavour to data interpretation. This is solely based on my learnings.	Yes the team does include authors from Aboriginal and racialised backgrounds. The team has published extensively on racism and health and is known internationally for its work in this area.